This workflow is shown in figure 16.4, and implemented in listing 16.1: this heuristic is called **random sampling**.

Listing 16.1 Random Sampling for Minimum Crossing Number Embedding

```
function randomEmbedding(graph, W, H, runs)
                            #1
  kBest ←
inf
#2
  for i in {1..runs} do
                  #3
    embedding ←
[]
   #4
    for j in {1..|graph.vertices|} do
                                    #5
      x ← random(0,
W)                                                          #6
      y ← random(0,
H)

    embedding.add((x,y))
        #7
    if not validateEmbedding(embedding, graph)
then                              #8
      i--
    else
      k ← edgeIntersections(graph,
embedding)                                              #9
      if k < kBest
then                                                          #10
        kBest ← k
        embeddingBest ← embedding
  return embeddingBest
                                                      #11
```